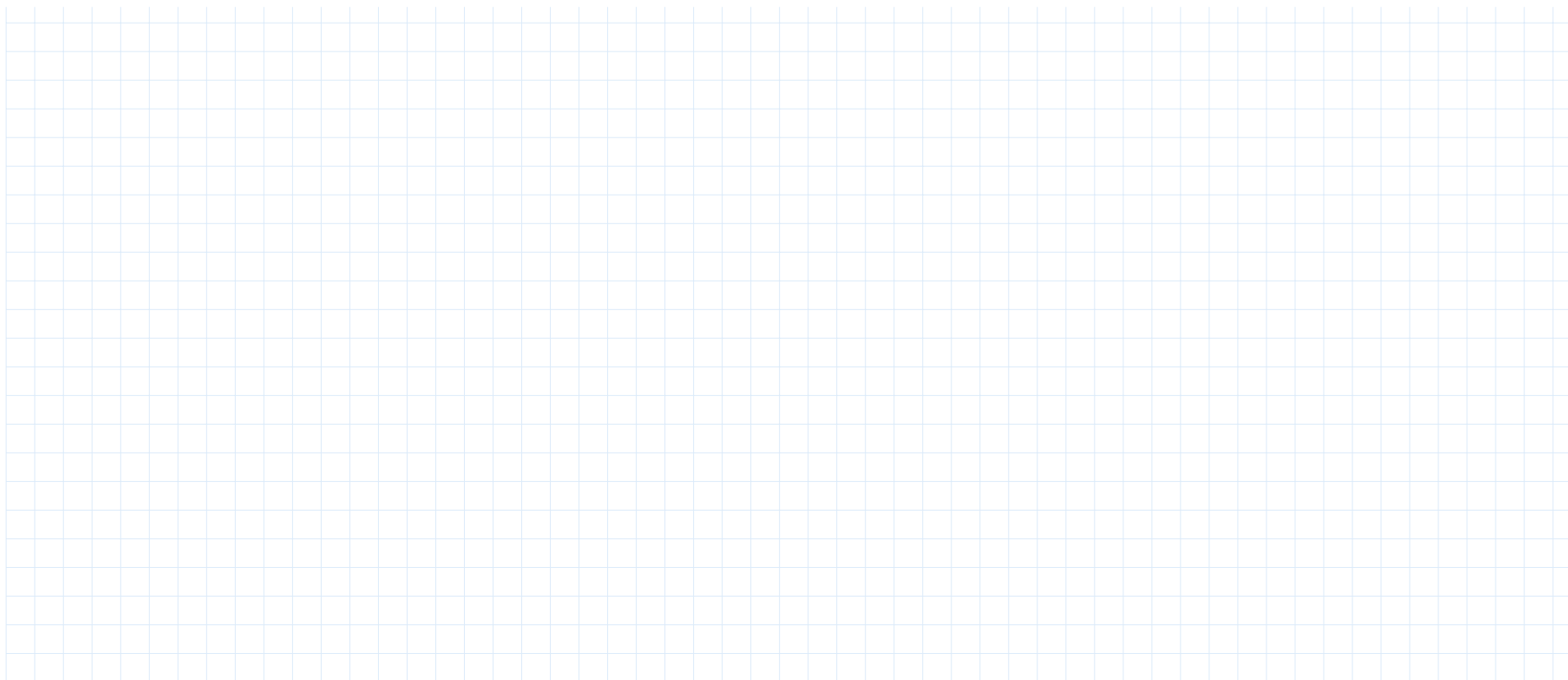


A point on the unit circle

On the unit circle $x^2 + y^2 = 1$, the point $P = (x, y)$ at angle θ gives

$$\cos \theta = x, \quad \sin \theta = y, \quad \tan \theta = \frac{y}{x} \ (x \neq 0).$$

Example. If $P = \left(-\frac{3}{5}, \frac{4}{5}\right)$, find $\sin \theta$, $\cos \theta$, and $\tan \theta$. Identify the quadrant.

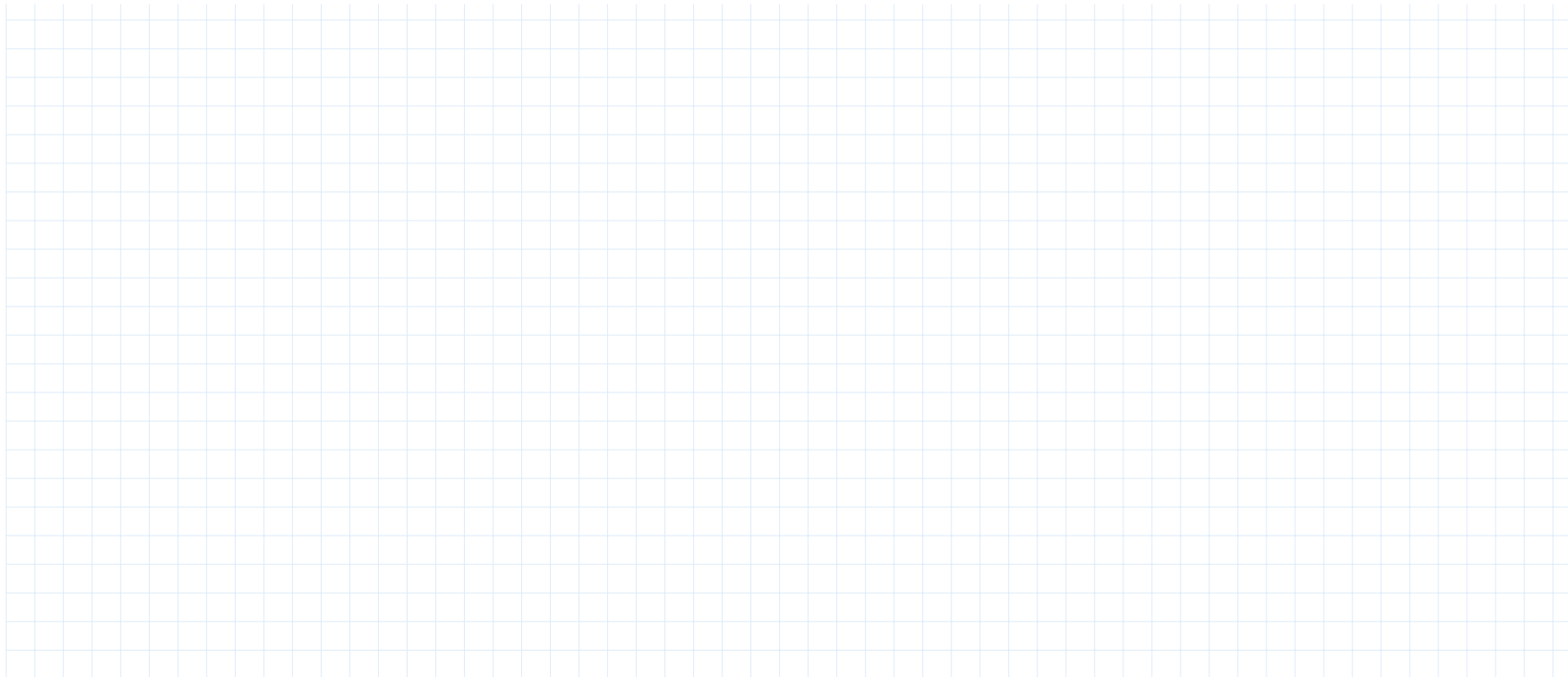


Reciprocals and undefined values

The remaining functions are

$$\csc \theta = \frac{1}{\sin \theta}, \quad \sec \theta = \frac{1}{\cos \theta}, \quad \cot \theta = \frac{\cos \theta}{\sin \theta}.$$

Example. Use the unit-circle points to find $\csc \pi$, $\sec \frac{\pi}{2}$, and $\cot \frac{\pi}{2}$. Explain each undefined value.



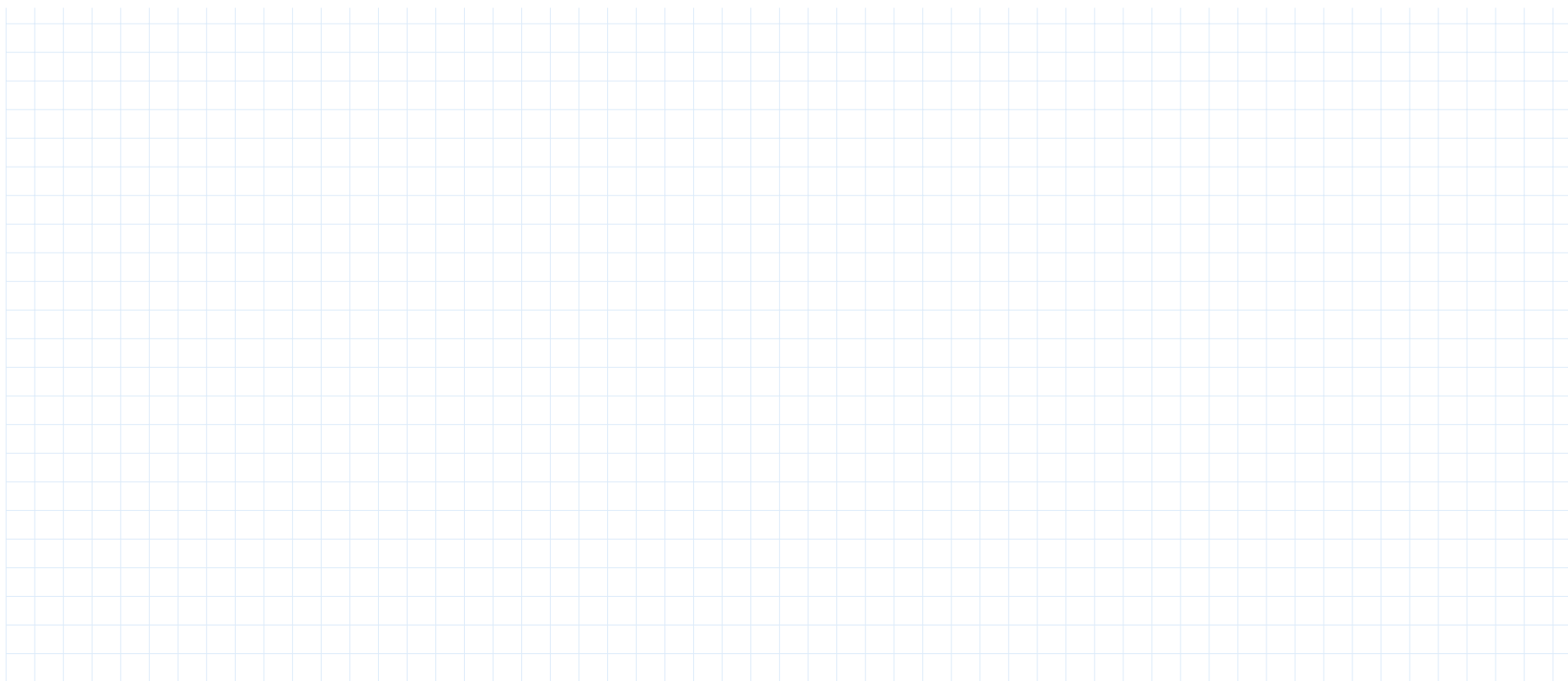
Special-angle values

The 45° - 45° - 90° and 30° - 60° - 90° triangles give the first-quadrant unit-circle coordinates.

Example. Build a table of $\sin \theta$ and $\cos \theta$ for

$$\theta = 0, \quad \frac{\pi}{6}, \quad \frac{\pi}{4}, \quad \frac{\pi}{3}, \quad \frac{\pi}{2}.$$

Show how a special triangle gives one of the non-axis points.

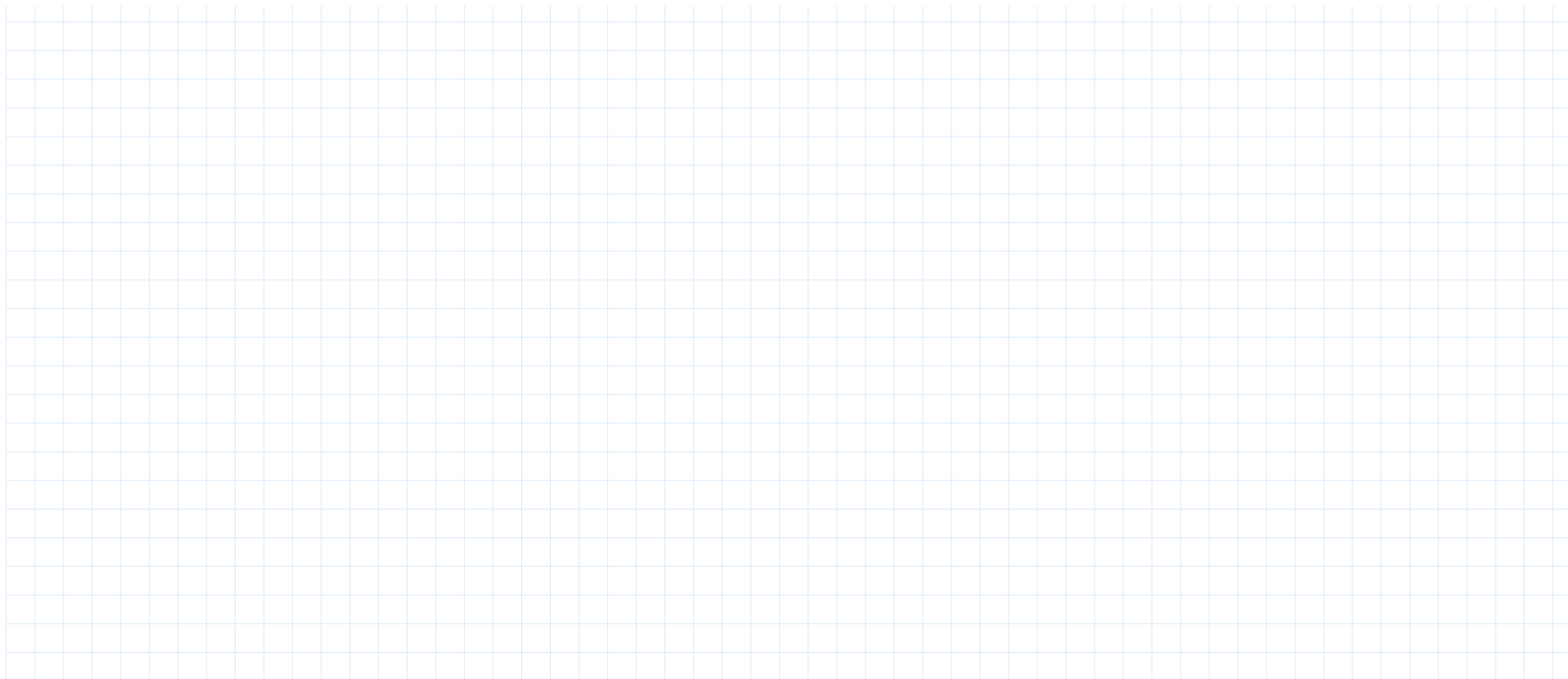


Beyond the first quadrant

Use the first-quadrant reference angle for the magnitude and the terminal point's quadrant for the sign.

Example. Find exact values, without a calculator:

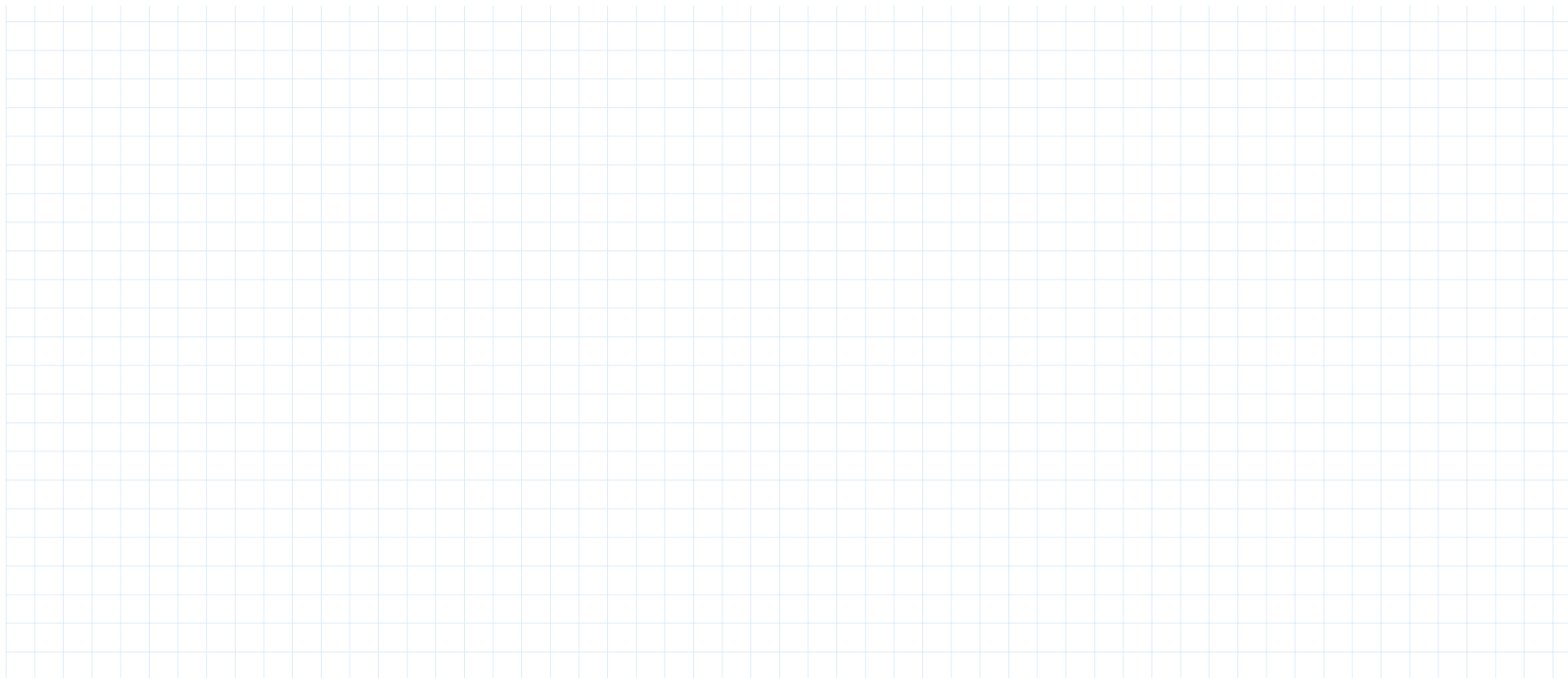
(a) $\sin \frac{5\pi}{6}$ (b) $\cos \frac{7\pi}{4}$ (c) $\tan \frac{4\pi}{3}$.



A circle of any radius

For a point (x, y) on the terminal side, let $r = \sqrt{x^2 + y^2} > 0$. Then $\sin \theta = \frac{y}{r}$ and $\cos \theta = \frac{x}{r}$.

Example. The point $(-8, -15)$ lies on the terminal side of θ . Find the exact values of all six trigonometric functions.



A degree symbol means degrees. An angle such as 1.2 or $\frac{5\pi}{18}$, with no degree symbol, is measured in radians.

Example. Approximate each value to four decimal places. State the calculator mode used.

(a) $\cos 38^\circ$ (b) $\sec(1.2)$ (c) $\tan \frac{5\pi}{18}$.

